E1 hand-writing workbook

75 real arXiv papers posted in 2025 (random, seed 0). For each paper, question **A** was extracted with the single fixed prompt `EXTRACT_RQ` (common/prompts.py); **edit it by hand** until it is faithful to the paper. Write every other field **yourself**. You need at least 60 complete items. Skip a paper if its question is unclear.

Rules (checked by `python E1/eval_contrast.py --check`)

- Jaccard(A, B) ≤ 0.3 · B_min keeps ≥ 75 % of A's words · Jaccard(A, C_tilde) ≤ 0.3
- ≥ 10 items of each near-miss type (the type is assigned on each page; 15 each)
- ≥ 10 items where **broader is longer than narrower** (marked ★ on the page)
- Jaccard = shared lower-case words / all distinct words of the two questions.

Near-miss types

role_swap	swap two roles: "how does X affect Y" → "how does Y affect X"
constraint_flip	flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"
goal_swap	change the goal: accuracy → speed, robustness → fairness
inquiry_change	change the kind of question: "how does" → "why does" / "can we" / "to what extent"
direction_flip	flip the direction: "improve" → "harm", "increase" → "reduce"

Worked example (from the paper)

A	How does data augmentation affect contrastive pre-training of image encoders without labelled data?
B	Without any annotations, what difference do random transformations of training pictures make to encoders that learn by contrasting views?
B_min	How does data augmentation influence contrastive pre-training of image encoders without labelled data?
C	How does contrastive pre-training of image encoders affect data augmentation without labelled data?
C_tilde	Without any annotations, what difference do networks that learn by contrasting views make to how training pictures are transformed?
narrower	How does colour-jitter augmentation affect contrastive pre-training of ResNet image encoders on ImageNet without labelled data?
broader	How do transformations of training examples change what self-supervised models learn, across data modalities, architectures and learning objectives?

After writing

1. Type your answers into `E1/contrast_template.jsonl` (same ids), save it as `E1/contrast.jsonl` (keep the h000 example line first).
2. Run the checker and fix what it flags.
3. **At least 24 hours later**, re-read every item and fix mistakes; note the date in NOTEBOOK.md.

h001 C type: role_swap ★ make broader LONGER than narrower

arXiv 2503.17882 · 2025-03-22

Think Before Refusal : Triggering Safety Reflection in LLMs to Mitigate False Refusal Behavior

Recent advancements in large language models (LLMs) have demonstrated that fine-tuning and human alignment can render LLMs harmless. In practice, such "harmlessness" behavior is mainly achieved by training models to reject harmful requests, such as "Explain how to burn down my neighbor's house", where the model appropriately declines to respond. However, this approach can inadvertently result in false refusal, where models reject benign queries as well, such as "Tell me how to kill a Python process". In this work, we demonstrate that prompting safety reflection before generating a response can mitigate false refusal behavior. Building on this finding, we introduce the Think-Before-Refusal (TBR) schema and conduct safety-aware instruction fine-tuning incorporating safety reflection. In an ablation study across 15 pre-trained models, we show that models fine-tuned with safety reflection significantly reduce false refusal behavior while maintaining safety and overall performance compared to those fine-tuned without safety reflection.

> **Extracted A:** What is the impact of incorporating safety reflection into safety-aware instruction fine-tuning for mitigating false refusal behavior in large language models?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

---

broader A question that contains A as a special case.

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h002 C type: constraint_flip

arXiv 2507.11049 · 2025-07-15

Journalism-Guided Agentic In-Context Learning for News Stance Detection

As online news consumption grows, personalized recommendation systems have become integral to digital journalism. However, these systems risk reinforcing filter bubbles and political polarization by failing to incorporate diverse perspectives. Stance detection -- identifying a text's position on a target -- can help mitigate this by enabling viewpoint-aware recommendations and data-driven analyses of media bias. Yet, existing stance detection research remains largely limited to short texts and high-resource languages. To address these gaps, we introduce \textsc{K-News-Stance}, the first Korean dataset for article-level stance detection, comprising 2,000 news articles with article-level and 21,650 segment-level stance annotations across 47 societal issues. We also propose \textsc{JoA-ICL}, a \textbf{Jo}urnalism-guided \textbf{A}gentic \textbf{I}n-\textbf{C}ontext \textbf{L}earning framework that employs a language model agent to predict the stances of key structural segments (e.g., leads, quotations), which are then aggregated to infer the overall article stance. Experiments showed that \textsc{JoA-ICL} outperforms existing stance detection methods, highlighting the benefits of segment-level agency in capturing the overall position of long-form news articles. Two case studies further demonstrate its broader utility in promoting viewpoint diversity in news recommendations and uncovering patterns of media bias.

Extracted A: What is the impact of incorporating diverse perspectives into stance detection algorithms?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

narrower A special case of A (add a dataset, model, domain...).

broader A question that contains A as a special case.

h003 C type: goal_swap

arXiv 2504.20752 · 2025-04-29

Grokking in the Wild: Data Augmentation for Real-World Multi-Hop Reasoning with Transformers

Transformers have achieved great success in numerous NLP tasks but continue to exhibit notable gaps in multi-step factual reasoning, especially when real-world knowledge is sparse. Recent advances in grokking have demonstrated that neural networks can transition from memorizing to perfectly generalizing once they detect underlying logical patterns - yet these studies have primarily used small, synthetic tasks. In this paper, for the first time, we extend grokking to real-world factual data and address the challenge of dataset sparsity by augmenting existing knowledge graphs with carefully designed synthetic data to raise the ratio ϕ_r of inferred facts to atomic facts above the threshold required for grokking. Surprisingly, we find that even factually incorrect synthetic data can strengthen emergent reasoning circuits rather than degrade accuracy, as it forces the model to rely on relational structure rather than memorization. When evaluated on multi-hop reasoning benchmarks, our approach achieves up to 95-100% accuracy on 2WikiMultiHopQA - substantially improving over strong baselines and matching or exceeding current state-of-the-art results. We further provide an in-depth analysis of how increasing ϕ_r drives the formation of generalizing circuits inside Transformers. Our findings suggest that grokking-based data augmentation can unlock implicit multi-hop reasoning capabilities, opening the door to more robust and interpretable factual reasoning in large-scale language models.

> **Extracted A:** How does increasing the ratio \(\phi_r\) affect the formation of generalizing circuits in Transformers for multi-hop reasoning?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h004 C type: inquiry_change

arXiv 2503.02199 · 2025-03-04

Words or Vision: Do Vision-Language Models Have Blind Faith in Text?

Vision-Language Models (VLMs) excel in integrating visual and textual information for vision-centric tasks, but their handling of inconsistencies between modalities is underexplored. We investigate VLMs' modality preferences when faced with visual data and varied textual inputs in vision-centered settings. By introducing textual variations to four vision-centric tasks and evaluating ten Vision-Language Models (VLMs), we discover a \emph{``blind faith in text"} phenomenon: VLMs disproportionately trust textual data over visual data when inconsistencies arise, leading to significant performance drops under corrupted text and raising safety concerns. We analyze factors influencing this text bias, including instruction prompts, language model size, text relevance, token order, and the interplay between visual and textual certainty. While certain factors, such as scaling up the language model size, slightly mitigate text bias, others like token order can exacerbate it due to positional biases inherited from language models. To address this issue, we explore supervised fine-tuning with text augmentation and demonstrate its effectiveness in reducing text bias. Additionally, we provide a theoretical analysis suggesting that the blind faith in text phenomenon may stem from an imbalance of pure text and multi-modal data during training. Our findings highlight the need for balanced training and careful consideration of modality interactions in VLMs to enhance their robustness and reliability in handling multi-modal data inconsistencies.

Extracted A: What factors influence the blind faith in text phenomenon in Vision-Language Models?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

---

broader A question that contains A as a special case.

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h005 C type: direction_flip

arXiv 2509.07622 · 2025-09-09

MaLei at MultiClinSUM: Summarisation of Clinical Documents using Perspective-Aware Iterative Self-Prompting with LLMs

Efficient communication between patients and clinicians plays an important role in shared decision-making. However, clinical reports are often lengthy and filled with clinical jargon, making it difficult for domain experts to identify important aspects in the document efficiently. This paper presents the methodology we applied in the MultiClinSUM shared task for summarising clinical case documents. We used an Iterative Self-Prompting technique on large language models (LLMs) by asking LLMs to generate task-specific prompts and refine them via example-based few-shot learning. Furthermore, we used lexical and embedding space metrics, ROUGE and BERT-score, to guide the model fine-tuning with epochs. Our submission using perspective-aware ISP on GPT-4 and GPT-4o achieved ROUGE scores (46.53, 24.68, 30.77) and BERTscores (87.84, 83.25, 85.46) for (P, R, F1) from the official evaluation on 3,396 clinical case reports from various specialties extracted from open journals. The high BERTscore indicates that the model produced semantically equivalent output summaries compared to the references, even though the overlap at the exact lexicon level is lower, as reflected in the lower ROUGE scores. This work sheds some light on how perspective-aware ISP (PA-ISP) can be deployed for clinical report summarisation and support better communication between patients and clinicians.

> **Extracted A:** What is the impact of perspective-aware iterative self-prompting on the quality of clinical case document summarization?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h006 C type: role_swap

arXiv 2510.18999 · 2025-10-21

OREN: Octree Residual Network for Real-Time Euclidean Signed Distance Mapping

Reconstructing signed distance functions (SDFs) from point cloud data benefits many robot autonomy capabilities, including localization, mapping, motion planning, and control. Methods that support online and large-scale SDF reconstruction often rely on discrete volumetric data structures, which affects the continuity and differentiability of the SDF estimates. Neural network methods have demonstrated high-fidelity differentiable SDF reconstruction but they tend to be less efficient, experience catastrophic forgetting and memory limitations in large environments, and are often restricted to truncated SDF. This work proposes OREN, a hybrid method that combines an explicit prior from octree interpolation with an implicit residual from neural network regression. Our method achieves non-truncated (Euclidean) SDF reconstruction with computational and memory efficiency comparable to volumetric methods and differentiability and accuracy comparable to neural network methods. Extensive experiments demonstrate that OREN outperforms the state of the art in terms of accuracy and efficiency, providing a scalable solution for downstream tasks in robotics and computer vision.

> **Extracted A:** What is the proposed hybrid method OREN for reconstructing signed distance functions from point clouds?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h007 C type: constraint_flip ★ make broader LONGER than narrower

arXiv 2510.21823 · 2025-10-21

Explainable Deep Learning in Medical Imaging: Brain Tumor and Pneumonia Detection

Deep Learning (DL) holds enormous potential for improving medical imaging diagnostics, yet the lack of interpretability in most models hampers clinical trust and adoption. This paper presents an explainable deep learning framework for detecting brain tumors in MRI scans and pneumonia in chest X-ray images using two leading Convolutional Neural Networks, ResNet50 and DenseNet121. These models were trained on publicly available Kaggle datasets comprising 7,023 brain MRI images and 5,863 chest X-ray images, achieving high classification performance. DenseNet121 consistently outperformed ResNet50 with 94.3 percent vs. 92.5 percent accuracy for brain tumors and 89.1 percent vs. 84.4 percent accuracy for pneumonia. For better explainability, Gradient-weighted Class Activation Mapping (Grad-CAM) was integrated to create heatmap visualizations superimposed on the test images, indicating the most influential image regions in the decision-making process. Interestingly, while both models produced accurate results, Grad-CAM showed that DenseNet121 consistently focused on core pathological regions, whereas ResNet50 sometimes scattered attention to peripheral or non-pathological areas. Combining deep learning and explainable AI offers a promising path toward reliable, interpretable, and clinically useful diagnostic tools.

> **Extracted A:** What is the relationship between the use of explainable deep learning in medical imaging and its ability to improve diagnostic accuracy?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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broader A question that contains A as a special case.

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h008 C type: goal_swap

arXiv 2501.06389 · 2025-01-10

Kolmogorov-Arnold networks for metal surface defect classification

This paper presents the application of Kolmogorov-Arnold Networks (KAN) in classifying metal surface defects. Specifically, steel surfaces are analyzed to detect defects such as cracks, inclusions, patches, pitted surfaces, and scratches. Drawing on the Kolmogorov-Arnold theorem, KAN provides a novel approach compared to conventional multilayer perceptrons (MLPs), facilitating more efficient function approximation by utilizing spline functions. The results show that KAN networks can achieve better accuracy than convolutional neural networks (CNNs) with fewer parameters, resulting in faster convergence and improved performance in image classification.

> **Extracted A:** What is the effectiveness of using Kolmogorov-Arnold Networks in metal surface defect classification?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h009 C type: inquiry_change

arXiv 2508.03092 · 2025-08-05

Toward Verifiable Misinformation Detection: A Multi-Tool LLM Agent Framework

With the proliferation of Large Language Models (LLMs), the detection of misinformation has become increasingly important and complex. This research proposes an innovative verifiable misinformation detection LLM agent that goes beyond traditional true/false binary judgments. The agent actively verifies claims through dynamic interaction with diverse web sources, assesses information source credibility, synthesizes evidence, and provides a complete verifiable reasoning process. Our designed agent architecture includes three core tools: precise web search tool, source credibility assessment tool and numerical claim verification tool. These tools enable the agent to execute multi-step verification strategies, maintain evidence logs, and form comprehensive assessment conclusions. We evaluate using standard misinformation datasets such as FakeNewsNet, comparing with traditional machine learning models and LLMs. Evaluation metrics include standard classification metrics, quality assessment of reasoning processes, and robustness testing against rewritten content. Experimental results show that our agent outperforms baseline methods in misinformation detection accuracy, reasoning transparency, and resistance to information rewriting, providing a new paradigm for trustworthy AI-assisted fact-checking.

Extracted A: What is the proposed innovative verifiable misinformation detection LLM agent framework?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h010 C type: direction_flip

arXiv 2505.00586 · 2025-05-01

ParkDiffusion: Heterogeneous Multi-Agent Multi-Modal Trajectory Prediction for Automated Parking using Diffusion Models

Automated parking is a critical feature of Advanced Driver Assistance Systems (ADAS), where accurate trajectory prediction is essential to bridge perception and planning modules. Despite its significance, research in this domain remains relatively limited, with most existing studies concentrating on single-modal trajectory prediction of vehicles. In this work, we propose ParkDiffusion, a novel approach that predicts the trajectories of both vehicles and pedestrians in automated parking scenarios. ParkDiffusion employs diffusion models to capture the inherent uncertainty and multi-modality of future trajectories, incorporating several key innovations. First, we propose a dual map encoder that processes soft semantic cues and hard geometric constraints using a two-step cross-attention mechanism. Second, we introduce an adaptive agent type embedding module, which dynamically conditions the prediction process on the distinct characteristics of vehicles and pedestrians. Third, to ensure kinematic feasibility, our model outputs control signals that are subsequently used within a kinematic framework to generate physically feasible trajectories. We evaluate ParkDiffusion on the Dragon Lake Parking (DLP) dataset and the Intersections Drone (inD) dataset. Our work establishes a new baseline for heterogeneous trajectory prediction in parking scenarios, outperforming existing methods by a considerable margin.

> **Extracted A:** What is the performance comparison between ParkDiffusion and existing methods in predicting heterogeneous multi-agent multi-modal trajectories in automated parking scenarios?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

h011 C type: role_swap

arXiv 2501.11847 · 2025-01-21

A Survey on Memory-Efficient Transformer-Based Model Training in AI for Science

Scientific research faces high costs and inefficiencies with traditional methods, but the rise of deep learning and large language models (LLMs) offers innovative solutions. This survey reviews transformer-based LLM applications across scientific fields such as biology, medicine, chemistry, and meteorology, underscoring their role in advancing research. However, the continuous expansion of model size has led to significant memory demands, hindering further development and application of LLMs for science. This survey systematically reviews and categorizes memory-efficient pre-training techniques for large-scale transformers, including algorithm-level, system-level, and hardware-software co-optimization. Using AlphaFold 2 as an example, we demonstrate how tailored memory optimization methods can reduce storage needs while preserving prediction accuracy. By bridging model efficiency and scientific application needs, we hope to provide insights for scalable and cost-effective LLM training in AI for science.

> **Extracted A:** How do tailored memory optimization methods improve scalability and cost-effectiveness in large-scale transformer-based LLM training for scientific applications?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h012 C type: constraint_flip

arXiv 2502.13723 · 2025-02-19

Direct Value Optimization: Improving Chain-of-Thought Reasoning in LLMs with Refined Values

We introduce Direct Value Optimization (DVO), an innovative reinforcement learning framework for enhancing large language models in complex reasoning tasks. Unlike traditional methods relying on preference labels, DVO utilizes value signals at individual reasoning steps, optimizing models via a mean squared error loss. The key benefit of DVO lies in its fine-grained supervision, circumventing the need for labor-intensive human annotations. Target values within the DVO are estimated using either Monte Carlo Tree Search or an outcome value model. Our empirical analysis on both mathematical and commonsense reasoning tasks shows that DVO consistently outperforms existing offline preference optimization techniques, even with fewer training steps. These findings underscore the importance of value signals in advancing reasoning capabilities and highlight DVO as a superior methodology under scenarios lacking explicit human preference information.

> **Extracted A:** What is the role of value signals in improving chain-of-thought reasoning in large language models?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h013 C type: goal_swap ★ make broader LONGER than narrower

arXiv 2504.15415 · 2025-04-21

IV-Bench: A Benchmark for Image-Grounded Video Perception and Reasoning in Multimodal LLMs

Existing evaluation frameworks for Multimodal Large Language Models (MLLMs) primarily focus on image reasoning or general video understanding tasks, largely overlooking the significant role of image context in video comprehension. To bridge this gap, we propose IV-Bench, the first comprehensive benchmark for evaluating Image-Grounded Video Perception and Reasoning. IV-Bench consists of 967 videos paired with 2,585 meticulously annotated image-text queries across 13 tasks (7 perception and 6 reasoning tasks) and 5 representative categories. Extensive evaluations of state-of-the-art open-source (e.g., InternVL2.5, Qwen2.5-VL) and closed-source (e.g., GPT-4o, Gemini2-Flash and Gemini2-Pro) MLLMs demonstrate that current models substantially underperform in image-grounded video Perception and Reasoning, merely achieving at most 28.9% accuracy. Further analysis reveals key factors influencing model performance on IV-Bench, including inference pattern, frame number, and resolution. Additionally, through a simple data synthesis approach, we demonstratethe challenges of IV- Bench extend beyond merely aligning the data format in the training proecss. These findings collectively provide valuable insights for future research. Our codes and data are released in https://github.com/multimodal-art-projection/IV-Bench.

Extracted A: What key factors influence model performance on the IV-Bench benchmark?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

narrower A special case of A (add a dataset, model, domain...).

broader A question that contains A as a special case.

h014 C type: inquiry_change

arXiv 2503.23925 · 2025-03-31

CoMatch: Dynamic Covisibility-Aware Transformer for Bilateral Subpixel-Level Semi-Dense Image Matching

This prospective study proposes CoMatch, a novel semi-dense image matcher with dynamic covisibility awareness and bilateral subpixel accuracy. Firstly, observing that modeling context interaction over the entire coarse feature map elicits highly redundant computation due to the neighboring representation similarity of tokens, a covisibility-guided token condenser is introduced to adaptively aggregate tokens in light of their covisibility scores that are dynamically estimated, thereby ensuring computational efficiency while improving the representational capacity of aggregated tokens simultaneously. Secondly, considering that feature interaction with massive non-covisible areas is distracting, which may degrade feature distinctiveness, a covisibility-assisted attention mechanism is deployed to selectively suppress irrelevant message broadcast from non-covisible reduced tokens, resulting in robust and compact attention to relevant rather than all ones. Thirdly, we find that at the fine-level stage, current methods adjust only the target view's keypoints to subpixel level, while those in the source view remain restricted at the coarse level and thus not informative enough, detrimental to keypoint location-sensitive usages. A simple yet potent fine correlation module is developed to refine the matching candidates in both source and target views to subpixel level, attaining attractive performance improvement. Thorough experimentation across an array of public benchmarks affirms CoMatch's promising accuracy, efficiency, and generalizability.

> **Extracted A:** What is the proposed method for improving the accuracy and efficiency of bilaterally subpixel-level semi-dense image matching using dynamic covisibility-aware token aggregation?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

---

broader A question that contains A as a special case.

h015 C type: direction_flip

arXiv 2510.12143 · 2025-10-14

Fairness-Constrained Optimization Attack in Federated Learning

Federated learning (FL) is a privacy-preserving machine learning technique that facilitates collaboration among participants across demographics. FL enables model sharing, while restricting the movement of data. Since FL provides participants with independence over their training data, it becomes susceptible to poisoning attacks. Such collaboration also propagates bias among the participants, even unintentionally, due to different data distribution or historical bias present in the data. This paper proposes an intentional fairness attack, where a client maliciously sends a biased model, by increasing the fairness loss while training, even considering homogeneous data distribution. The fairness loss is calculated by solving an optimization problem for fairness metrics such as demographic parity and equalized odds. The attack is insidious and hard to detect, as it maintains global accuracy even after increasing the bias. We evaluate our attack against the state-of-the-art Byzantine-robust and fairness-aware aggregation schemes over different datasets, in various settings. The empirical results demonstrate the attack efficacy by increasing the bias up to 90\%, even in the presence of a single malicious client in the FL system.

Extracted A: What is the proposed fairness-constrained optimization attack on federated learning?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h016 C type: role_swap

arXiv 2501.11407 · 2025-01-20

A Truly Sparse and General Implementation of Gradient-Based Synaptic Plasticity

Online synaptic plasticity rules derived from gradient descent achieve high accuracy on a wide range of practical tasks. However, their software implementation often requires tediously hand-derived gradients or using gradient backpropagation which sacrifices the online capability of the rules. In this work, we present a custom automatic differentiation (AD) pipeline for sparse and online implementation of gradient-based synaptic plasticity rules that generalizes to arbitrary neuron models. Our work combines the programming ease of backpropagation-type methods for forward AD while being memory-efficient. To achieve this, we exploit the advantageous compute and memory scaling of online synaptic plasticity by providing an inherently sparse implementation of AD where expensive tensor contractions are replaced with simple element-wise multiplications if the tensors are diagonal. Gradient-based synaptic plasticity rules such as eligibility propagation (e-prop) have exactly this property and thus profit immensely from this feature. We demonstrate the alignment of our gradients with respect to gradient backpropagation on an synthetic task where e-prop gradients are exact, as well as audio speech classification benchmarks. We demonstrate how memory utilization scales with network size without dependence on the sequence length, as expected from forward AD methods.

> **Extracted A:** What is the optimal method for implementing sparse and online gradient-based synaptic plasticity rules?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h017 C type: constraint_flip

arXiv 2508.11943 · 2025-08-16

Learning Marked Temporal Point Process Explanations based on Counterfactual and Factual Reasoning

Neural network-based Marked Temporal Point Process (MTPP) models have been widely adopted to model event sequences in high-stakes applications, raising concerns about the trustworthiness of outputs from these models. This study focuses on Explanation for MTPP, aiming to identify the minimal and rational explanation, that is, the minimum subset of events in history, based on which the prediction accuracy of MTPP matches that based on full history to a great extent and better than that based on the complement of the subset. This study finds that directly defining Explanation for MTPP as counterfactual explanation or factual explanation can result in irrational explanations. To address this issue, we define Explanation for MTPP as a combination of counterfactual explanation and factual explanation. This study proposes Counterfactual and Factual Explainer for MTPP (CFF) to solve Explanation for MTPP with a series of deliberately designed techniques. Experiments demonstrate the correctness and superiority of CFF over baselines regarding explanation quality and processing efficiency.

> **Extracted A:** What is the correct way to explain an event sequence using neural network-based Marked Temporal Point Process models?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h018 C type: goal_swap

arXiv 2509.21262 · 2025-09-25

Un-Doubling Diffusion: LLM-guided Disambiguation of Homonym Duplication

Homonyms are words with identical spelling but distinct meanings, which pose challenges for many generative models. When a homonym appears in a prompt, diffusion models may generate multiple senses of the word simultaneously, which is known as homonym duplication. This issue is further complicated by an Anglocentric bias, which includes an additional translation step before the text-to-image model pipeline. As a result, even words that are not homonymous in the original language may become homonyms and lose their meaning after translation into English. In this paper, we introduce a method for measuring duplication rates and conduct evaluations of different diffusion models using both automatic evaluation utilizing Vision-Language Models (VLM) and human evaluation. Additionally, we investigate methods to mitigate the homonym duplication problem through prompt expansion, demonstrating that this approach also effectively reduces duplication related to Anglocentric bias. The code for the automatic evaluation pipeline is publicly available.

> **Extracted A:** What measures can be taken to reduce duplication related to Anglocentric bias in diffusion models?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h019 C type: inquiry_change ★ make broader LONGER than narrower

arXiv 2503.14337 · 2025-03-18

PENCIL: Long Thoughts with Short Memory

While state-of-the-art LLMs have demonstrated great promise of using long Chains-of-Thought (CoT) to boost reasoning, scaling it up to more challenging problems at test-time is fundamentally limited by suboptimal memory usage -- intermediate computations accumulate indefinitely in context even when no longer needed for future thoughts. We introduce PENCIL, which incorporates a novel reduction mechanism into the autoregressive generation process that recursively cleans up intermediate thoughts based on patterns learned from training. By iteratively generating and erasing thoughts, PENCIL can think deeper to solve harder problems using shorter context and less compute. Empirically, we observe PENCIL is significantly more effective and efficient than CoT. For example, we demonstrate PENCIL with a small 25M-parameter transformer and 2048 context length solves Einstein's puzzle -- a task that challenges much larger models like GPT-4. Theoretically, we prove PENCIL can perform universal efficient computation by simulating any Turing machines with optimal time and space complexity, and thus can solve arbitrary computable tasks that are otherwise intractable for vanilla CoT.

> **Extracted A:** What is the effect of incorporating a novel reduction mechanism into the autoregressive generation process on the efficiency and effectiveness of thinking deeper to solve harder problems?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

---

broader A question that contains A as a special case.

---

h020 C type: direction_flip

arXiv 2503.02835 · 2025-03-04

In-Depth Analysis of Automated Acne Disease Recognition and Classification

Facial acne is a common disease, especially among adolescents, negatively affecting both physically and psychologically. Classifying acne is vital to providing the appropriate treatment. Traditional visual inspection or expert scanning is time-consuming and difficult to differentiate acne types. This paper introduces an automated expert system for acne recognition and classification. The proposed method employs a machine learning-based technique to classify and evaluate six types of acne diseases to facilitate the diagnosis of dermatologists. The pre-processing phase includes contrast improvement, smoothing filter, and RGB to L*a*b color conversion to eliminate noise and improve the classification accuracy. Then, a clustering-based segmentation method, k-means clustering, is applied for segmenting the disease-affected regions that pass through the feature extraction step. Characteristics of these disease-affected regions are extracted based on a combination of gray-level co-occurrence matrix (GLCM) and Statistical features. Finally, five different machine learning classifiers are employed to classify acne diseases. Experimental results show that the Random Forest (RF) achieves the highest accuracy of 98.50%, which is promising compared to the state-of-the-art methods.

> **Extracted A:** What is the performance of the Random Forest classifier in classifying acne diseases?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

---

broader A question that contains A as a special case.

---

h021 C type: role_swap

arXiv 2502.15732 · 2025-02-05

Data Wrangling Task Automation Using Code-Generating Language Models

Ensuring data quality in large tabular datasets is a critical challenge, typically addressed through data wrangling tasks. Traditional statistical methods, though efficient, cannot often understand the semantic context and deep learning approaches are resource-intensive, requiring task and dataset-specific training. To overcome these shortcomings, we present an automated system that utilizes large language models to generate executable code for tasks like missing value imputation, error detection, and error correction. Our system aims to identify inherent patterns in the data while leveraging external knowledge, effectively addressing both memory-dependent and memory-independent tasks.

> **Extracted A:** What is the role of code-generation language models in ensuring data quality in large tabular datasets?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

---

broader A question that contains A as a special case.

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h022 C type: constraint_flip

arXiv 2506.05982 · 2025-06-06

MCA-Bench: A Multimodal Benchmark for Evaluating CAPTCHA Robustness Against VLM-based Attacks

As automated attack techniques rapidly advance, CAPTCHAs remain a critical defense mechanism against malicious bots. However, existing CAPTCHA schemes encompass a diverse range of modalities -- from static distorted text and obfuscated images to interactive clicks, sliding puzzles, and logic-based questions -- yet the community still lacks a unified, large-scale, multimodal benchmark to rigorously evaluate their security robustness. To address this gap, we introduce MCA-Bench, a comprehensive and reproducible benchmarking suite that integrates heterogeneous CAPTCHA types into a single evaluation protocol. Leveraging a shared vision-language model backbone, we fine-tune specialized cracking agents for each CAPTCHA category, enabling consistent, cross-modal assessments. Extensive experiments reveal that MCA-Bench effectively maps the vulnerability spectrum of modern CAPTCHA designs under varied attack settings, and crucially offers the first quantitative analysis of how challenge complexity, interaction depth, and model solvability interrelate. Based on these findings, we propose three actionable design principles and identify key open challenges, laying the groundwork for systematic CAPTCHA hardening, fair benchmarking, and broader community collaboration. Datasets and code are available online.

> **Extracted A:** What is the proposed approach for evaluating the security robustness of CAPTCHAs using a multimodal benchmark?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h023 C type: goal_swap

arXiv 2508.05513 · 2025-08-07

Streamlining Admission with LOR Insights: AI-Based Leadership Assessment in Online Master's Program

Letters of recommendation (LORs) provide valuable insights into candidates' capabilities and experiences beyond standardized test scores. However, reviewing these text-heavy materials is time-consuming and labor-intensive. To address this challenge and support the admission committee in providing feedback for students' professional growth, our study introduces LORI: LOR Insights, a novel AI-based detection tool for assessing leadership skills in LORs submitted by online master's program applicants. By employing natural language processing and leveraging large language models using RoBERTa and LLAMA, we seek to identify leadership attributes such as teamwork, communication, and innovation. Our latest RoBERTa model achieves a weighted F1 score of 91.6%, a precision of 92.4%, and a recall of 91.6%, showing a strong level of consistency in our test data. With the growing importance of leadership skills in the STEM sector, integrating LORI into the graduate admissions process is crucial for accurately assessing applicants' leadership capabilities. This approach not only streamlines the admissions process but also automates and ensures a more comprehensive evaluation of candidates' capabilities.

> **Extracted A:** What is the impact of incorporating Natural Language Processing (NLP) and Large Language Models (LLMs) like RoBERTa and LLAMA on the accuracy and reliability of identifying leadership attributes from letters of recommendation submitted by online master's program applicants?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h024 C type: inquiry_change

arXiv 2505.14825 · 2025-05-20

Assimilative Causal Inference

Causal inference is fundamental across scientific disciplines, yet existing methods struggle to capture instantaneous, time-evolving causal relationships in complex, high-dimensional systems. In this paper, assimilative causal inference (ACI) is developed, which is a methodological framework that leverages Bayesian data assimilation to trace causes backward from observed effects. ACI solves the inverse problem rather than quantifying forward influence. It uniquely identifies dynamic causal interactions without requiring observations of candidate causes, accommodates short datasets, and, in principle, can be implemented in high-dimensional settings by employing efficient data assimilation algorithms. Crucially, it provides online tracking of causal roles that may reverse intermittently and facilitates a mathematically rigorous criterion for the causal influence range, revealing how far effects propagate. The effectiveness of ACI is demonstrated by complex dynamical systems showcasing intermittency and extreme events. ACI opens valuable pathways for studying complex systems, where transient causal structures are critical.

> **Extracted A:** What is the unique approach of assimilative causal inference (ACI) in solving the inverse problem of causality?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h025 C type: direction_flip ★ make broader LONGER than narrower

arXiv 2510.18817 · 2025-10-21

Fine-Tuned Thoughts: Leveraging Chain-of-Thought Reasoning for Industrial Asset Health Monitoring

Small Language Models (SLMs) are becoming increasingly popular in specialized fields, such as industrial applications, due to their efficiency, lower computational requirements, and ability to be fine-tuned for domain-specific tasks, enabling accurate and cost-effective solutions. However, performing complex reasoning using SLMs in specialized fields such as Industry 4.0 remains challenging. In this paper, we propose a knowledge distillation framework for industrial asset health, which transfers reasoning capabilities via Chain-of-Thought (CoT) distillation from Large Language Models (LLMs) to smaller, more efficient models (SLMs). We discuss the advantages and the process of distilling LLMs using multi-choice question answering (MCQA) prompts to enhance reasoning and refine decision-making. We also perform in-context learning to verify the quality of the generated knowledge and benchmark the performance of fine-tuned SLMs with generated knowledge against widely used LLMs. The results show that the fine-tuned SLMs with CoT reasoning outperform the base models by a significant margin, narrowing the gap to their LLM counterparts. Our code is open-sourced at: https://github.com/IBM/FailureSensorIQ.

> **Extracted A:** What are the benefits of using Chain-of-Thought reasoning in industrial asset health monitoring compared to traditional methods?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h026 C type: role_swap ★ make broader LONGER than narrower

arXiv 2510.09658 · 2025-10-07

Gradient-Sign Masking for Task Vector Transport Across Pre-Trained Models

When a new release of a foundation model is published, practitioners typically need to repeat fine-tuning, even if the same task was already tackled in the previous version. A promising alternative is to reuse the parameter changes (i.e., task vectors) that capture how a model adapts to a specific task. However, these vectors often fail to transfer across different pre-trained models because their parameter spaces are misaligned. In this work, we show that successful transfer depends strongly on the gradient-sign structure of the new model. Based on this insight, we propose GradFix, which approximates the ideal sign structure and leverages it to transfer knowledge using only a handful of labeled samples. Notably, this requires no additional fine-tuning: we only compute a few target-model gradients without parameter updates and mask the source task vector accordingly. This yields an update that is locally aligned with the target loss landscape, effectively rebasing the task vector onto the new pre-training. We provide a theoretical guarantee that our method ensures first-order descent. Empirically, we demonstrate significant performance gains on vision and language benchmarks, consistently outperforming naive task vector addition and few-shot fine-tuning. We further show that transporting task vectors improves multi-task and multi-source model merging. Code is available at https://github.com/fillo-rinaldi/GradFix.

> **Extracted A:** What is the optimal approach for transferring task vectors across pre-trained models based on the gradient-sign structure?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h027 C type: constraint_flip

arXiv 2505.03783 · 2025-04-30

A general physics-constrained method for the modelling of equation's closure terms with sparse data

Accurate modeling of closure terms is a critical challenge in engineering and scientific research, particularly when data is sparse (scarse or incomplete), making widely applicable models difficult to develop. This study proposes a novel approach for constructing closure models in such challenging scenarios. We introduce a Series-Parallel Multi-Network Architecture that integrates Physics-Informed Neural Networks (PINNs) to incorporate physical constraints and heterogeneous data from multiple initial and boundary conditions, while employing dedicated subnetworks to independently model unknown closure terms, enhancing generalizability across diverse problems. These closure models are integrated into an accurate Partial Differential Equation (PDE) solver, enabling robust solutions to complex predictive simulations in engineering applications.

> **Extracted A:** What is the proposed approach for constructing closure models in sparse data scenarios using Physics-Informed Neural Networks?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain…).

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broader A question that contains A as a special case.

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h028 C type: goal_swap

arXiv 2506.09457 · 2025-06-11

Towards Bridging the Reward-Generation Gap in Direct Alignment Algorithms

Direct Alignment Algorithms (DAAs), such as Direct Preference Optimization (DPO) and Simple Preference Optimization (SimPO), have emerged as efficient alternatives to Reinforcement Learning from Human Feedback (RLHF) algorithms for aligning large language models (LLMs) with human preferences. However, DAAs suffer from a fundamental limitation we identify as the "reward-generation gap", a discrepancy between training objectives and autoregressive decoding dynamics. In this paper, we consider that one contributor to the reward-generation gap is the mismatch between the inherent importance of prefix tokens during the LLM generation process and how this importance is reflected in the implicit reward functions of DAAs. To bridge the gap, we adopt a token-level MDP perspective of DAAs to analyze its limitations and introduce a simple yet effective approach called Prefix-Oriented Equal-length Training (POET), which truncates both preferred and dispreferred responses to match the shorter one's length. We conduct experiments with DPO and SimPO, two representative DAAs, demonstrating that POET improves over their standard implementations, achieving up to 11.8 points in AlpacaEval 2 and overall improvements across downstream tasks. These results underscore the need to mitigate the reward-generation gap in DAAs by better aligning training objectives with autoregressive decoding dynamics.

> **Extracted A:** What is the impact of the mismatch between the inherent importance of prefix tokens and how it affects the effectiveness of Direct Alignment Algorithms (DAAs)?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h029 C type: inquiry_change

arXiv 2506.17558 · 2025-06-21

SynDaCaTE: A Synthetic Dataset For Evaluating Part-Whole Hierarchical Inference

Learning to infer object representations, and in particular part-whole hierarchies, has been the focus of extensive research in computer vision, in pursuit of improving data efficiency, systematic generalisation, and robustness. Models which are \emph{designed} to infer part-whole hierarchies, often referred to as capsule networks, are typically trained end-to-end on supervised tasks such as object classification, in which case it is difficult to evaluate whether such a model \emph{actually} learns to infer part-whole hierarchies, as claimed. To address this difficulty, we present a SYNthetic DAtaset for CApsule Testing and Evaluation, abbreviated as SynDaCaTE, and establish its utility by (1) demonstrating the precise bottleneck in a prominent existing capsule model, and (2) demonstrating that permutation-equivariant self-attention is highly effective for parts-to-wholes inference, which motivates future directions for designing effective inductive biases for computer vision.

> **Extracted A:** What is the specific bottleneck in a prominent existing capsule model and how does permutation-equivariant self-attention contribute to effectively learning part-whole hierarchies?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h030 C type: direction_flip

arXiv 2504.08740 · 2025-03-23

Recommendation System in Advertising and Streaming Media: Unsupervised Data Enhancement Sequence Suggestions

Sequential recommendation is an extensively explored approach to capturing users' evolving preferences based on past interactions, aimed at predicting their next likely choice. Despite significant advancements in this domain, including methods based on RNNs and self-attention, challenges like limited supervised signals and noisy data caused by unintentional clicks persist. To address these challenges, some studies have incorporated unsupervised learning by leveraging local item contexts within individual sequences. However, these methods often overlook the intricate associations between items across multiple sequences and are susceptible to noise in item co-occurrence patterns. In this context, we introduce a novel framework, Global Unsupervised Data-Augmentation (UDA4SR), which adopts a graph contrastive learning perspective to generate more robust item embeddings for sequential recommendation. Our approach begins by integrating Generative Adversarial Networks (GANs) for data augmentation, which serves as the first step to enhance the diversity and richness of the training data. Then, we build a Global Item Relationship Graph (GIG) based on all user interaction sequences. Subsequently, we employ graph contrastive learning on the refined graph to enhance item embeddings by capturing complex global associations. To model users' dynamic and diverse interests more effectively, we enhance the CapsNet module with a novel target-attention mechanism. Extensive experiments show that UDA4SR significantly outperforms state-of-the-art approaches.

> **Extracted A:** What is the effectiveness of Global Unsupervised Data-Augmentation (UDA4SR) in enhancing item embeddings for sequential recommendation?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

h031 C type: role_swap

arXiv 2503.23502 · 2025-03-30

Boosting Omnidirectional Stereo Matching with a Pre-trained Depth Foundation Model

Omnidirectional depth perception is essential for mobile robotics applications that require scene understanding across a full 360{\deg} field of view. Camera-based setups offer a cost-effective option by using stereo depth estimation to generate dense, high-resolution depth maps without relying on expensive active sensing. However, existing omnidirectional stereo matching approaches achieve only limited depth accuracy across diverse environments, depth ranges, and lighting conditions, due to the scarcity of real-world data. We present DFI-OmniStereo, a novel omnidirectional stereo matching method that leverages a large-scale pre-trained foundation model for relative monocular depth estimation within an iterative optimization-based stereo matching architecture. We introduce a dedicated two-stage training strategy to utilize the relative monocular depth features for our omnidirectional stereo matching before scale-invariant fine-tuning. DFI-OmniStereo achieves state-of-the-art results on the real-world Helvipad dataset, reducing disparity MAE by approximately 16% compared to the previous best omnidirectional stereo method.

> **Extracted A:** What is the key contribution of DFI-OmniStereo in boosting omnidirectional stereo matching performance?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h032 C type: constraint_flip ★ make broader LONGER than narrower

arXiv 2501.01472 · 2025-01-01

Augmented Contrastive Clustering with Uncertainty-Aware Prototyping for Time Series Test Time Adaptation

Test-time adaptation aims to adapt pre-trained deep neural networks using solely online unlabelled test data during inference. Although TTA has shown promise in visual applications, its potential in time series contexts remains largely unexplored. Existing TTA methods, originally designed for visual tasks, may not effectively handle the complex temporal dynamics of real-world time series data, resulting in suboptimal adaptation performance. To address this gap, we propose Augmented Contrastive Clustering with Uncertainty-aware Prototyping (ACCUP), a straightforward yet effective TTA method for time series data. Initially, our approach employs augmentation ensemble on the time series data to capture diverse temporal information and variations, incorporating uncertainty-aware prototypes to distill essential characteristics. Additionally, we introduce an entropy comparison scheme to selectively acquire more confident predictions, enhancing the reliability of pseudo labels. Furthermore, we utilize augmented contrastive clustering to enhance feature discriminability and mitigate error accumulation from noisy pseudo labels, promoting cohesive clustering within the same class while facilitating clear separation between different classes. Extensive experiments conducted on three real-world time series datasets and an additional visual dataset demonstrate the effectiveness and generalization potential of the proposed method, advancing the underexplored realm of TTA for time series data.

> **Extracted A:** What is the impact of incorporating uncertainty-aware prototypes into Augmented Contrastive Clustering with Uncertainty-Aware Prototyping for time series test-time adaptation?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h033 C type: goal_swap

arXiv 2502.16357 · 2025-02-22

LegalBench.PT: A Benchmark for Portuguese Law

The recent application of LLMs to the legal field has spurred the creation of benchmarks across various jurisdictions and languages. However, no benchmark has yet been specifically designed for the Portuguese legal system. In this work, we present LegalBench.PT, the first comprehensive legal benchmark covering key areas of Portuguese law. To develop LegalBench.PT, we first collect long-form questions and answers from real law exams, and then use GPT-4o to convert them into multiple-choice, true/false, and matching formats. Once generated, the questions are filtered and processed to improve the quality of the dataset. To ensure accuracy and relevance, we validate our approach by having a legal professional review a sample of the generated questions. Although the questions are synthetically generated, we show that their basis in human-created exams and our rigorous filtering and processing methods applied result in a reliable benchmark for assessing LLMs' legal knowledge and reasoning abilities. Finally, we evaluate the performance of leading LLMs on LegalBench.PT and investigate potential biases in GPT-4o's responses. We also assess the performance of Portuguese lawyers on a sample of questions to establish a baseline for model comparison and validate the benchmark.

Extracted A: What is the first comprehensive legal benchmark covering key areas of Portuguese law?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h034 C type: inquiry_change

arXiv 2505.17860 · 2025-05-23

Multi-Person Interaction Generation from Two-Person Motion Priors

Generating realistic human motion with high-level controls is a crucial task for social understanding, robotics, and animation. With high-quality MOCAP data becoming more available recently, a wide range of data-driven approaches have been presented. However, modelling multi-person interactions still remains a less explored area. In this paper, we present Graph-driven Interaction Sampling, a method that can generate realistic and diverse multi-person interactions by leveraging existing two-person motion diffusion models as motion priors. Instead of training a new model specific to multi-person interaction synthesis, our key insight is to spatially and temporally separate complex multi-person interactions into a graph structure of two-person interactions, which we name the Pairwise Interaction Graph. We thus decompose the generation task into simultaneous single-person motion generation conditioned on one other's motion. In addition, to reduce artifacts such as interpenetrations of body parts in generated multi-person interactions, we introduce two graph-dependent guidance terms into the diffusion sampling scheme. Unlike previous work, our method can produce various high-quality multi-person interactions without having repetitive individual motions. Extensive experiments demonstrate that our approach consistently outperforms existing methods in reducing artifacts when generating a wide range of two-person and multi-person interactions.

> **Extracted A:** What is the key insight behind Graph-driven Interaction Sampling (GIDS) in generating realistic multi-person interactions?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h035 C type: direction_flip

arXiv 2501.04682 · 2025-01-08

Towards System 2 Reasoning in LLMs: Learning How to Think With Meta Chain-of-Thought

We propose a novel framework, Meta Chain-of-Thought (Meta-CoT), which extends traditional Chain-of-Thought (CoT) by explicitly modeling the underlying reasoning required to arrive at a particular CoT. We present empirical evidence from state-of-the-art models exhibiting behaviors consistent with in-context search, and explore methods for producing Meta-CoT via process supervision, synthetic data generation, and search algorithms. Finally, we outline a concrete pipeline for training a model to produce Meta-CoTs, incorporating instruction tuning with linearized search traces and reinforcement learning post-training. Finally, we discuss open research questions, including scaling laws, verifier roles, and the potential for discovering novel reasoning algorithms. This work provides a theoretical and practical roadmap to enable Meta-CoT in LLMs, paving the way for more powerful and human-like reasoning in artificial intelligence.

> **Extracted A:** What is the proposed framework Meta Chain-of-Thought (Meta-CoT) for extending traditional Chain-of-Thought?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h036 C type: role_swap

arXiv 2505.20653 · 2025-05-27

RoGA: Towards Generalizable Deepfake Detection through Robust Gradient Alignment

Recent advancements in domain generalization for deepfake detection have attracted significant attention, with previous methods often incorporating additional modules to prevent overfitting to domain-specific patterns. However, such regularization can hinder the optimization of the empirical risk minimization (ERM) objective, ultimately degrading model performance. In this paper, we propose a novel learning objective that aligns generalization gradient updates with ERM gradient updates. The key innovation is the application of perturbations to model parameters, aligning the ascending points across domains, which specifically enhances the robustness of deepfake detection models to domain shifts. This approach effectively preserves domain-invariant features while managing domain-specific characteristics, without introducing additional regularization. Experimental results on multiple challenging deepfake detection datasets demonstrate that our gradient alignment strategy outperforms state-of-the-art domain generalization techniques, confirming the efficacy of our method. The code is available at https://github.com/Lynn0925/RoGA.

> **Extracted A:** What is the key innovation in the proposed learning objective for generalizable deepfake detection?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

---

broader A question that contains A as a special case.

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h037 C type: constraint_flip

arXiv 2504.05350 · 2025-04-06

Non-linear Phillips Curve for India: Evidence from Explainable Machine Learning

The conventional linear Phillips curve model, while widely used in policymaking, often struggles to deliver accurate forecasts in the presence of structural breaks and inherent nonlinearities. This paper addresses these limitations by leveraging machine learning methods within a New Keynesian Phillips Curve framework to forecast and explain headline inflation in India, a major emerging economy. Our analysis demonstrates that machine learning-based approaches significantly outperform standard linear models in forecasting accuracy. Moreover, by employing explainable machine learning techniques, we reveal that the Phillips curve relationship in India is highly nonlinear, characterized by thresholds and interaction effects among key variables. Headline inflation is primarily driven by inflation expectations, followed by past inflation and the output gap, while supply shocks, except rainfall, exert only a marginal influence. These findings highlight the ability of machine learning models to improve forecast accuracy and uncover complex, nonlinear dynamics in inflation data, offering valuable insights for policymakers.

> **Extracted A:** What are the significant differences between traditional linear Phillips curves and the new Keynesian Phillips curve framework when applied to Indian inflation data?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

---

broader A question that contains A as a special case.

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h038 C type: goal_swap ★ make broader LONGER than narrower

arXiv 2506.22845 · 2025-06-28

Quantum Neural Networks for Wind Energy Forecasting: A Comparative Study of Performance and Scalability with Classical Models

Quantum Neural Networks (QNNs), a prominent approach in Quantum Machine Learning (QML), are emerging as a powerful alternative to classical machine learning methods. Recent studies have focused on the applicability of QNNs to various tasks, such as time-series forecasting, prediction, and classification, across a wide range of applications, including cybersecurity and medical imaging. With the increased use of smart grids driven by the integration of renewable energy systems, machine learning plays an important role in predicting power demand and detecting system disturbances. This study provides an in-depth investigation of QNNs for predicting the power output of a wind turbine. We assess the predictive performance and simulation time of six QNN configurations that are based on the Z Feature Map for data encoding and varying ansatz structures. Through detailed cross-validation experiments and tests on an unseen hold-out dataset, we experimentally demonstrate that QNNs can achieve predictive performance that is competitive with, and in some cases marginally better than, the benchmarked classical approaches. Our results also reveal the effects of dataset size and circuit complexity on predictive performance and simulation time. We believe our findings will offer valuable insights for researchers in the energy domain who wish to incorporate quantum machine learning into their work.

> **Extracted A:** What are the effects of dataset size and circuit complexity on predictive performance and simulation time when using quantum neural networks for wind energy forecasting?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h039 C type: inquiry_change

arXiv 2502.20350 · 2025-02-27

KEDRec-LM: A Knowledge-distilled Explainable Drug Recommendation Large Language Model

Drug discovery is a critical task in biomedical natural language processing (NLP), yet explainable drug discovery remains underexplored. Meanwhile, large language models (LLMs) have shown remarkable abilities in natural language understanding and generation. Leveraging LLMs for explainable drug discovery has the potential to improve downstream tasks and real-world applications. In this study, we utilize open-source drug knowledge graphs, clinical trial data, and PubMed publications to construct a comprehensive dataset for the explainable drug discovery task, named \textbf{expRxRec}. Furthermore, we introduce \textbf{KEDRec-LM}, an instruction-tuned LLM which distills knowledge from rich medical knowledge corpus for drug recommendation and rationale generation. To encourage further research in this area, we will publicly release\footnote{A copy is attached with this submission} both the dataset and KEDRec-LM.

> **Extracted A:** What is the potential impact of leveraging large language models for explainable drug discovery on improving downstream tasks and real-world applications?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h040 C type: direction_flip

arXiv 2505.10438 · 2025-05-15

Koopman Eigenfunction-Based Identification and Optimal Nonlinear Control of Turbojet Engine

Gas turbine engines are complex and highly nonlinear dynamical systems. Deriving their physics-based models can be challenging because it requires performance characteristics that are not always available, often leading to many simplifying assumptions. This paper discusses the limitations of conventional experimental methods used to derive component-level and locally linear parameter-varying models, and addresses these issues by employing identification techniques based on data collected from standard engine operation under closed-loop control. The rotor dynamics are estimated using the sparse identification of nonlinear dynamics. Subsequently, the autonomous part of the dynamics is mapped into an optimally constructed Koopman eigenfunction space. This process involves eigenvalue optimization using metaheuristic algorithms and temporal projection, followed by gradient-based eigenfunction identification. The resulting Koopman model is validated against an in-house reference component-level model. A globally optimal nonlinear feedback controller and a Kalman estimator are then designed within the eigenfunction space and compared to traditional and gain-scheduled proportional-integral controllers, as well as a proposed internal model control approach. The eigenmode structure enables targeting individual modes during optimization, leading to improved performance tuning. Results demonstrate that the Koopman-based controller surpasses other benchmark controllers in both reference tracking and disturbance rejection under sea-level and varying flight conditions, due to its global nature.

> **Extracted A:** What is the effectiveness of the Koopman-based controller in improving performance tuning for turbojet engine dynamics?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

h041 C type: role_swap

arXiv 2510.10546 · 2025-10-12

GLOFNet -- A Multimodal Dataset for GLOF Monitoring and Prediction

Glacial Lake Outburst Floods (GLOFs) are rare but destructive hazards in high mountain regions, yet predictive research is hindered by fragmented and unimodal data. Most prior efforts emphasize post-event mapping, whereas forecasting requires harmonized datasets that combine visual indicators with physical precursors. We present GLOFNet, a multimodal dataset for GLOF monitoring and prediction, focused on the Shisper Glacier in the Karakoram. It integrates three complementary sources: Sentinel-2 multispectral imagery for spatial monitoring, NASA ITS_LIVE velocity products for glacier kinematics, and MODIS Land Surface Temperature records spanning over two decades. Preprocessing included cloud masking, quality filtering, normalization, temporal interpolation, augmentation, and cyclical encoding, followed by harmonization across modalities. Exploratory analysis reveals seasonal glacier velocity cycles, long-term warming of ~0.8 K per decade, and spatial heterogeneity in cryospheric conditions. The resulting dataset, GLOFNet, is publicly available to support future research in glacial hazard prediction. By addressing challenges such as class imbalance, cloud contamination, and coarse resolution, GLOFNet provides a structured foundation for benchmarking multimodal deep learning approaches to rare hazard prediction.

> **Extracted A:** What are the key challenges faced in predicting glacial lake outburst floods using multimodal datasets?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h042 C type: constraint_flip

arXiv 2510.12117 · 2025-10-14

Locket: Robust Feature-Locking Technique for Language Models

Chatbot service providers (e.g., OpenAI) rely on tiered subscription plans to generate revenue, offering black-box access to basic models for free users and advanced models to paying subscribers. However, this approach is unprofitable and inflexible. A pay-to-unlock scheme for premium features (e.g., math, coding) offers a more sustainable alternative. Enabling such a scheme requires a feature-locking technique (FLoTE) that is (i) effective in refusing locked features, (ii) utility-preserving for unlocked features, (iii) robust against evasion or unauthorized credential sharing, and (iv) scalable to multiple features and clients. Existing FLoTEs (e.g., password-locked models) fail to meet these criteria. To fill this gap, we present Locket, the first robust and scalable FLoTE to enable pay-to-unlock schemes. We develop a framework for adversarial training and merging of feature-locking adapters, which enables Locket to selectively disable specific features of a model. Evaluation shows that Locket is effective (100% refusal rate), utility-preserving (≤ 7% utility degradation), robust (≤ 5% attack success rate), and scalable to multiple features and clients.

> **Extracted A:** What is the effectiveness of Locket, a robust and scalable feature-locking technique for language models?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h043 C type: goal_swap

arXiv 2505.12043 · 2025-05-17

MoL for LLMs: Dual-Loss Optimization to Enhance Domain Expertise While Preserving General Capabilities

Although large language models (LLMs) perform well in general tasks, domain-specific applications suffer from hallucinations and accuracy limitations. Continual Pre-Training (CPT) approaches encounter two key issues: (1) domain-biased data degrades general language skills, and (2) improper corpus-mixture ratios limit effective adaptation. To address these, we propose a novel framework, Mixture of Losses (MoL), which decouples optimization objectives for domain-specific and general corpora. Specifically, cross-entropy (CE) loss is applied to domain-corpus to ensure knowledge acquisition, while Kullback-Leibler (KL) divergence aligns general-corpus training with the base model's foundational capabilities. This dual-loss architecture preserves universal skills while enhancing domain expertise, avoiding catastrophic forgetting. Empirically, we validate that a 1:1 domain-to-general corpus ratio optimally balances training and overfitting without the need for extensive tuning or resource-intensive experiments. Furthermore, our experiments demonstrate significant performance gains compared to traditional CPT approaches, which often suffer from degradation in general language capabilities; our model achieves 27.9% higher accuracy on the Math-500 benchmark in the non-think reasoning mode, and an impressive 83.3% improvement on the challenging AIME25 subset in the think mode, underscoring the effectiveness of our approach.

> **Extracted A:** What is the optimal balance between training and overfitting when using dual-loss optimization for continuous pre-training of large language models?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h044 C type: inquiry_change ★ make broader LONGER than narrower

arXiv 2502.09198 · 2025-02-13

Understanding High-Dimensional Bayesian Optimization

Recent work reported that simple Bayesian optimization (BO) methods perform well for high-dimensional real-world tasks, seemingly contradicting prior work and tribal knowledge. This paper investigates why. We identify underlying challenges that arise in high-dimensional BO and explain why recent methods succeed. Our empirical analysis shows that vanishing gradients caused by Gaussian process (GP) initialization schemes play a major role in the failures of high-dimensional Bayesian optimization (HDBO) and that methods that promote local search behaviors are better suited for the task. We find that maximum likelihood estimation (MLE) of GP length scales suffices for state-of-the-art performance. Based on this, we propose a simple variant of MLE called MSR that leverages these findings to achieve state-of-the-art performance on a comprehensive set of real-world applications. We present targeted experiments to illustrate and confirm our findings.

Extracted A: Why do recent methods fail in high-dimensional Bayesian optimization?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h045 C type: direction_flip

arXiv 2502.20793 · 2025-02-28

Towards Ultimate NMR Resolution with Deep Learning

In multidimensional NMR spectroscopy, practical resolution is defined as the ability to distinguish and accurately determine signal positions against a background of overlapping peaks, thermal noise, and spectral artifacts. In the pursuit of ultimate resolution, we introduce Peak Probability Presentations (P^3)- a statistical spectral representation that assigns a probability to each spectral point, indicating the likelihood of a peak maximum occurring at that location. The mapping between the spectrum and P^3 is achieved using MR-Ai, a physics-inspired deep learning neural network architecture, designed to handle multidimensional NMR spectra. Furthermore, we demonstrate that MR-Ai enables coprocessing of multiple spectra, facilitating direct information exchange between datasets. This feature significantly enhances spectral quality, particularly in cases of highly sparse sampling. Performance of MR-Ai and high value of the P^3 are demonstrated on the synthetic data and spectra of Tau, MATL1, Calmodulin, and several other proteins.

> **Extracted A:** What is the relationship between the performance of MR-Ai and its ability to enhance spectral quality in multidimensional NMR spectroscopy?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h046 C type: role_swap

arXiv 2503.00371 · 2025-03-01

Jointly Understand Your Command and Intention:Reciprocal Co-Evolution between Scene-Aware 3D Human Motion Synthesis and Analysis

As two intimate reciprocal tasks, scene-aware human motion synthesis and analysis require a joint understanding between multiple modalities, including 3D body motions, 3D scenes, and textual descriptions. In this paper, we integrate these two paired processes into a Co-Evolving Synthesis-Analysis (CESA) pipeline and mutually benefit their learning. Specifically, scene-aware text-to-human synthesis generates diverse indoor motion samples from the same textual description to enrich human-scene interaction intra-class diversity, thus significantly benefiting training a robust human motion analysis system. Reciprocally, human motion analysis would enforce semantic scrutiny on each synthesized motion sample to ensure its semantic consistency with the given textual description, thus improving realistic motion synthesis. Considering that real-world indoor human motions are goal-oriented and path-guided, we propose a cascaded generation strategy that factorizes text-driven scene-specific human motion generation into three stages: goal inferring, path planning, and pose synthesizing. Coupling CESA with this powerful cascaded motion synthesis model, we jointly improve realistic human motion synthesis and robust human motion analysis in 3D scenes.

> **Extracted A:** What is the proposed approach for jointly understanding and enhancing the realism of human motion synthesis and analysis in 3D scenes?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h047 C type: constraint_flip

arXiv 2506.00743 · 2025-05-31

Assortment of Attention Heads: Accelerating Federated PEFT with Head Pruning and Strategic Client Selection

Parameter Efficient Fine-Tuning (PEFT) has become the de-facto approach in adapting Large Language Models (LLMs) for downstream tasks in Natural Language Processing. However, its adoption in privacy-preserving distributed learning frameworks, such as Federated Learning (FL), remains relatively limited. This is mainly due to challenges specific to FL, such as resource-constrained devices and diverse data distributions among clients. In this paper, we propose an efficient method to perform PEFT within the FL framework for Multi-Head Attention (MHA) based language models. We address the challenges through head pruning, a novel head-specific weighted aggregation mechanism, and a client selection strategy. Head pruning minimizes training complexity within the clients, guided by the importance score computed based on the confidence of the attention head. Weighted aggregation of heads ensures the global model captures crucial updates from diverse clients complementing our client selection strategy. We show results on the MultiNLI benchmark along with 20 Newsgroups, XL-Sum, and E2E NLG datasets. We use the MultiNLI dataset and T5-small model with LoRA as our PEFT method, attaining sparsity levels of up to 90%, resulting in a communication advantage of up to 1.8x and a reduction in training OPs of 3.9x while maintaining the accuracy drop under 2%.

> **Extracted A:** What is the proposed method for accelerating federated multi-head attention using head pruning and strategic client selection?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

---

broader A question that contains A as a special case.

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h048 C type: goal_swap

arXiv 2503.07120 · 2025-03-10

FEB-Cache: Frequency-Guided Exposure Bias Reduction for Enhancing Diffusion Transformer Caching

Diffusion Transformer (DiT) has exhibited impressive generation capabilities but faces great challenges due to its high computational complexity. To address this issue, various methods, notably feature caching, have been introduced. However, these approaches focus on aligning non-cache diffusion without analyzing why caching damage the generation processes. In this paper, we first confirm that the cache greatly amplifies the exposure bias, resulting in a decline in the generation quality. However, directly applying noise scaling is challenging for this issue due to the non-smoothness of exposure bias. We found that this phenomenon stems from the mismatch between its frequency response characteristics and the simple cache of Attention and MLP. Since these two components exhibit unique preferences for frequency signals, which provides us with a caching strategy to separate Attention and MLP to achieve an enhanced fit of exposure bias and reduce it. Based on this, we introduced FEB-Cache, a joint caching strategy that aligns with the non-exposed bias diffusion process (which gives us a higher performance cap) of caching Attention and MLP based on the frequency-guided cache table. Our approach combines a comprehensive understanding of the caching mechanism and offers a new perspective on leveraging caching to accelerate the diffusion process. Empirical results indicate that FEB-Cache optimizes model performance while concurrently facilitating acceleration. Code is available at https://github.com/aSleepyTree/EB-Cache.

> **Extracted A:** What is the impact of frequency-guided exposure bias reduction on the generation quality of Diffusion Transformers?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

h049 C type: inquiry_change

arXiv 2508.13236 · 2025-08-18

Uncertainty-Aware Learning Policy for Reliable Pulmonary Nodule Detection on Chest X-Ray

Early detection and rapid intervention of lung cancer are crucial. Nonetheless, ensuring an accurate diagnosis is challenging, as physicians' ability to interpret chest X-rays varies significantly depending on their experience and degree of fatigue. Although medical AI has been rapidly advancing to assist in diagnosis, physicians' trust in such systems remains limited, preventing widespread clinical adoption. This skepticism fundamentally stems from concerns about its diagnostic uncertainty. In clinical diagnosis, physicians utilize extensive background knowledge and clinical experience. In contrast, medical AI primarily relies on repetitive learning of the target lesion to generate diagnoses based solely on that data. In other words, medical AI does not possess sufficient knowledge to render a diagnosis, leading to diagnostic uncertainty. Thus, this study suggests an Uncertainty-Aware Learning Policy that can address the issue of knowledge deficiency by learning the physicians' background knowledge alongside the Chest X-ray lesion information. We used 2,517 lesion-free images and 656 nodule images, all obtained from Ajou University Hospital. The proposed model attained 92% (IoU 0.2 / FPPI 2) with a 10% enhancement in sensitivity compared to the baseline model while also decreasing entropy as a measure of uncertainty by 0.2.

> **Extracted A:** What is the impact of medical AI's reliance on repetitive learning of chest X-ray lesions on its ability to accurately diagnose pulmonary nodules?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

---

broader A question that contains A as a special case.

---

h050 C type: direction_flip ★ make broader LONGER than narrower

arXiv 2507.13325 · 2025-07-17

Social and Political Framing in Search Engine Results

Search engines play a crucial role in shaping public discourse by influencing how information is accessed and framed. While prior research has extensively examined various dimensions of search bias -- such as content prioritization, indexical bias, political polarization, and sources of bias -- an important question remains underexplored: how do search engines and ideologically-motivated user queries contribute to bias in search results. This study analyzes the outputs of major search engines using a dataset of political and social topics. The findings reveal that search engines not only prioritize content in ways that reflect underlying biases but also that ideologically-driven user queries exacerbate these biases, resulting in the amplification of specific narratives. Moreover, significant differences were observed across search engines in terms of the sources they prioritize. These results suggest that search engines may play a pivotal role in shaping public perceptions by reinforcing ideological divides, thereby contributing to the broader issue of information polarization.

Extracted A: How does search engine output influence the framing of political and social topics?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

narrower A special case of A (add a dataset, model, domain...).

broader A question that contains A as a special case.

h051 C type: role_swap ★ make broader LONGER than narrower

arXiv 2508.12243 · 2025-08-17

SEA-BED: How Do Embedding Models Represent Southeast Asian Languages?

Multilingual text embeddings are often assumed to encode meaning in a perspective-independent semantic space, yielding stable similarity judgments across tasks and languages. Our results show that this assumption does not hold in practice. We introduce SEA-BED, a large-scale benchmark covering 10 Southeast Asian (SEA) languages and diverse embedding tasks, designed to systematically examine how embedding performance varies across tasks, languages, and language-task combinations. Across extensive evaluations, we observe that no single model performs uniformly well across SEA languages; task difficulty differs markedly within languages, and success on one task does not reliably generalize to others. Language-task analyses further reveal highly non-uniform performance landscapes, where performance varies across different language-task combinations. These findings call for closer attention to performance measurements that provide an expansive view across languages and tasks to uncover inconsistencies in semantic representation. Based on these observations, we provide insights for future model development, including data, algorithmic, and architectural considerations.

> **Extracted A:** How do embedding models represent Southeast Asian languages differently from other languages?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

narrower A special case of A (add a dataset, model, domain...).

broader A question that contains A as a special case.

h052 C type: constraint_flip

arXiv 2509.26443 · 2025-09-30

Stabilization of nonlinear systems with unknown delays via delay-adaptive neural operator approximate predictors

This work establishes the first rigorous stability guarantees for approximate predictors in delay-adaptive control of nonlinear systems, addressing a key challenge in practical implementations where exact predictors are unavailable. We analyze two scenarios: (i) when the actuated input is directly measurable, and (ii) when it is estimated online. For the measurable input case, we prove semi-global practical asymptotic stability with an explicit bound proportional to the approximation error ϵ. For the unmeasured input case, we demonstrate local practical asymptotic stability, with the region of attraction explicitly dependent on both the initial delay estimate and the predictor approximation error. To bridge theory and practice, we show that neural operators-a flexible class of neural network-based approximators-can achieve arbitrarily small approximation errors, thus satisfying the conditions of our stability theorems. Numerical experiments on two nonlinear benchmark systems-a biological protein activator/repressor model and a micro-organism growth Chemostat model-validate our theoretical results. In particular, our numerical simulations confirm stability under approximate predictors, highlight the strong generalization capabilities of neural operators, and demonstrate a substantial computational speedup of up to 15x compared to a baseline fixed-point method.

> **Extracted A:** What are the conditions under which approximate predictors can be used to stabilize nonlinear systems with unknown delays?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain…).

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broader A question that contains A as a special case.

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h053 C type: goal_swap

arXiv 2509.02983 · 2025-09-03

DUViN: Diffusion-Based Underwater Visual Navigation via Knowledge-Transferred Depth Features

Autonomous underwater navigation remains a challenging problem due to limited sensing capabilities and the difficulty of constructing accurate maps in underwater environments. In this paper, we propose a Diffusion-based Underwater Visual Navigation policy via knowledge-transferred depth features, named DUViN, which enables vision-based end-to-end 4-DoF motion control for underwater vehicles in unknown environments. DUViN guides the vehicle to avoid obstacles and maintain a safe and perception awareness altitude relative to the terrain without relying on pre-built maps. To address the difficulty of collecting large-scale underwater navigation datasets, we propose a method that ensures robust generalization under domain shifts from in-air to underwater environments by leveraging depth features and introducing a novel model transfer strategy. Specifically, our training framework consists of two phases: we first train the diffusion-based visual navigation policy on in-air datasets using a pre-trained depth feature extractor. Secondly, we retrain the extractor on an underwater depth estimation task and integrate the adapted extractor into the trained navigation policy from the first step. Experiments in both simulated and real-world underwater environments demonstrate the effectiveness and generalization of our approach. The experimental videos are available at https://www.youtube.com/playlist?list=PLqt2s-RyCf1gfXJgFzKjmwIqYhrP4I-7Y.

> **Extracted A:** What is the proposed method for enabling vision-based end-to-end 4-DoF motion control in autonomous underwater vehicles through knowledge-transfer of depth features?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

h054 C type: inquiry_change

arXiv 2510.09072 · 2025-10-10

Emotion-Disentangled Embedding Alignment for Noise-Robust and Cross-Corpus Speech Emotion Recognition

Effectiveness of speech emotion recognition in real-world scenarios is often hindered by noisy environments and variability across datasets. This paper introduces a two-step approach to enhance the robustness and generalization of speech emotion recognition models through improved representation learning. First, our model employs EDRL (Emotion-Disentangled Representation Learning) to extract class-specific discriminative features while preserving shared similarities across emotion categories. Next, MEA (Multiblock Embedding Alignment) refines these representations by projecting them into a joint discriminative latent subspace that maximizes covariance with the original speech input. The learned EDRL-MEA embeddings are subsequently used to train an emotion classifier using clean samples from publicly available datasets, and are evaluated on unseen noisy and cross-corpus speech samples. Improved performance under these challenging conditions demonstrates the effectiveness of the proposed method.

> **Extracted A:** What is the effect of noise and variability on speech emotion recognition?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

---

broader A question that contains A as a special case.

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h055 C type: direction_flip

arXiv 2506.21463 · 2025-06-26

Aligning Spoken Dialogue Models from User Interactions

We propose a novel preference alignment framework for improving spoken dialogue models on real-time conversations from user interactions. Current preference learning methods primarily focus on text-based language models, and are not directly suited to the complexities of real-time speech interactions, with richer dynamics (e.g. interruption, interjection) and no explicit segmentation between speaker turns.We create a large-scale dataset of more than 150,000 preference pairs from raw multi-turn speech conversations, annotated with AI feedback, to cover preferences over both linguistic content and temporal context variations. We leverage offline alignment methods to finetune a full-duplex autoregressive speech-to-speech model. Extensive experiments demonstrate that feedback on generic conversations can be consistently effective in improving spoken dialogue models to produce more factual, safer and more contextually aligned interactions. We deploy the finetuned model and conduct holistic human evaluations to assess the impact beyond single-turn conversations. Our findings shed light on the importance of a well-calibrated balance among various dynamics, crucial for natural real-time speech dialogue systems.

> **Extracted A:** What is the importance of a well-calibrated balance among various dynamics for natural real-time speech dialogue systems?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

narrower A special case of A (add a dataset, model, domain...).

broader A question that contains A as a special case.

h056 C type: role_swap

arXiv 2506.11110 · 2025-06-08

AssertBench: A Benchmark for Evaluating Self-Assertion in Large Language Models

Recent benchmarks have probed factual consistency and rhetorical robustness in Large Language Models (LLMs). However, a knowledge gap exists regarding how directional framing of factually true statements influences model agreement, a common scenario for LLM users. AssertBench addresses this by sampling evidence-supported facts from FEVEROUS, a fact verification dataset. For each (evidence-backed) fact, we construct two framing prompts: one where the user claims the statement is factually correct, and another where the user claims it is incorrect. We then record the model's agreement and reasoning. The desired outcome is that the model asserts itself, maintaining consistent truth evaluation across both framings, rather than switching its evaluation to agree with the user. AssertBench isolates framing-induced variability from the model's underlying factual knowledge by stratifying results based on the model's accuracy on the same claims when presented neutrally. In doing so, this benchmark aims to measure an LLM's ability to "stick to its guns" when presented with contradictory user assertions about the same fact. The complete source code is available at https://github.com/achowd32/assert-bench.

> **Extracted A:** How does the isolation of framing-induced variability affect the model's ability to maintain consistent truth evaluation when presented with contradictory user assertions about the same fact?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h057 C type: constraint_flip ★ make broader LONGER than narrower

arXiv 2502.15168 · 2025-02-21

mStyleDistance: Multilingual Style Embeddings and their Evaluation

Style embeddings are useful for stylistic analysis and style transfer; however, only English style embeddings have been made available. We introduce Multilingual StyleDistance (mStyleDistance), a multilingual style embedding model trained using synthetic data and contrastive learning. We train the model on data from nine languages and create a multilingual STEL-or-Content benchmark (Wegmann et al., 2022) that serves to assess the embeddings' quality. We also employ our embeddings in an authorship verification task involving different languages. Our results show that mStyleDistance embeddings outperform existing models on these multilingual style benchmarks and generalize well to unseen features and languages. We make our model publicly available at https://huggingface.co/StyleDistance/mstyledistance .

> **Extracted A:** What is the performance of Multilingual StyleDistance embeddings compared to existing models on the Wiegmann et al., 2022 multilingual style benchmark?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h058 C type: goal_swap

arXiv 2505.20793 · 2025-05-27

Rendering-Aware Reinforcement Learning for Vector Graphics Generation

Scalable Vector Graphics (SVG) offer a powerful format for representing visual designs as interpretable code. Recent advances in vision-language models (VLMs) have enabled high-quality SVG generation by framing the problem as a code generation task and leveraging large-scale pretraining. VLMs are particularly suitable for this task as they capture both global semantics and fine-grained visual patterns, while transferring knowledge across vision, natural language, and code domains. However, existing VLM approaches often struggle to produce faithful and efficient SVGs because they never observe the rendered images during training. Although differentiable rendering for autoregressive SVG code generation remains unavailable, rendered outputs can still be compared to original inputs, enabling evaluative feedback suitable for reinforcement learning (RL). We introduce RLRF (Reinforcement Learning from Rendering Feedback), an RL method that enhances SVG generation in autoregressive VLMs by leveraging feedback from rendered SVG outputs. Given an input image, the model generates SVG roll-outs that are rendered and compared to the original image to compute a reward. This visual fidelity feedback guides the model toward producing more accurate, efficient, and semantically coherent SVGs. RLRF significantly outperforms supervised fine-tuning, addressing common failure modes and enabling precise, high-quality SVG generation with strong structural understanding and generalization.

> **Extracted A:** What is the impact of using rendered outputs in comparison to original inputs for evaluating SVG generation in autoregressive VLMs?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h059 C type: inquiry_change

arXiv 2501.10866 · 2025-01-18

QGAPHEnsemble : Combining Hybrid QLSTM Network Ensemble via Adaptive Weighting for Short Term Weather Forecasting

Accurate weather forecasting holds significant importance, serving as a crucial tool for decision-making in various industrial sectors. The limitations of statistical models, assuming independence among data points, highlight the need for advanced methodologies. The correlation between meteorological variables necessitate models capable of capturing complex dependencies. This research highlights the practical efficacy of employing advanced machine learning techniques proposing GenHybQLSTM and BO-QEnsemble architecture based on adaptive weight adjustment strategy. Through comprehensive hyper-parameter optimization using hybrid quantum genetic particle swarm optimisation algorithm and Bayesian Optimization, our model demonstrates a substantial improvement in the accuracy and reliability of meteorological predictions through the assessment of performance metrics such as MSE (Mean Squared Error) and MAPE (Mean Absolute Percentage Prediction Error). The paper highlights the importance of optimized ensemble techniques to improve the performance the given weather forecasting task.

> **Extracted A:** What is the optimal combination of hybrid QLSTM network ensemble and adaptive weight adjustment strategy for improving the accuracy and reliability of short-term weather forecasting?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h060 C type: direction_flip

arXiv 2507.14847 · 2025-07-20

Time-Aware Attention for Enhanced Electronic Health Records Modeling

Electronic Health Records (EHR) contain valuable clinical information for predicting patient outcomes and guiding healthcare decisions. However, effectively modeling Electronic Health Records (EHRs) requires addressing data heterogeneity and complex temporal patterns. Standard approaches often struggle with irregular time intervals between clinical events. We propose TALE-EHR, a Transformer-based framework featuring a novel time-aware attention mechanism that explicitly models continuous temporal gaps to capture fine-grained sequence dynamics. To complement this temporal modeling with robust semantics, TALE-EHR leverages embeddings derived from standardized code descriptions using a pre-trained Large Language Model (LLM), providing a strong foundation for understanding clinical concepts. Experiments on the MIMIC-IV and PIC dataset demonstrate that our approach outperforms state-of-the-art baselines on tasks such as disease progression forecasting. TALE-EHR underscores the benefit of integrating explicit, continuous temporal modeling with strong semantic representations provides a powerful solution for advancing EHR analysis.

> **Extracted A:** What is the benefit of integrating explicit, continuous temporal modeling with strong semantic representations in enhancing electronic health records modeling?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h061 C type: role_swap

arXiv 2508.12257 · 2025-08-17

Structuring the Unstructured: A Systematic Review of Text-to-Structure Generation for Agentic AI with a Universal Evaluation Framework

The evolution of AI systems toward agentic operation and context-aware retrieval necessitates transforming unstructured text into structured formats like tables, knowledge graphs, and charts. While such conversions enable critical applications from summarization to data mining, current research lacks a comprehensive synthesis of methodologies, datasets, and metrics. This systematic review examines text-to-structure techniques and the encountered challenges, evaluates current datasets and assessment criteria, and outlines potential directions for future research. We also introduce a universal evaluation framework for structured outputs, establishing text-to-structure as foundational infrastructure for next-generation AI systems.

> **Extracted A:** What are the current limitations and challenges in text-to-structure generation for agentic AI?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

narrower A special case of A (add a dataset, model, domain...).

broader A question that contains A as a special case.

h062 C type: constraint_flip

arXiv 2509.02075 · 2025-09-02

How Instruction-Tuning Imparts Length Control: A Cross-Lingual Mechanistic Analysis

Adhering to explicit length constraints, such as generating text with a precise word count, remains a significant challenge for Large Language Models (LLMs). This study aims at investigating the differences between foundation models and their instruction-tuned counterparts, on length-controlled text generation in English and Italian. We analyze both performance and internal component contributions using Cumulative Weighted Attribution, a metric derived from Direct Logit Attribution. Our findings reveal that instruction-tuning substantially improves length control, primarily by specializing components in deeper model layers. Specifically, attention heads in later layers of IT models show increasingly positive contributions, particularly in English. In Italian, while attention contributions are more attenuated, final-layer MLPs exhibit a stronger positive role, suggesting a compensatory mechanism. These results indicate that instruction-tuning reconfigures later layers for task adherence, with component-level strategies potentially adapting to linguistic context.

> **Extracted A:** How does instruction-tuning improve length control in LLMs?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h063 C type: goal_swap ★ make broader LONGER than narrower

arXiv 2503.11088 · 2025-03-14

Multi-View Industrial Anomaly Detection with Epipolar Constrained Cross-View Fusion

Multi-camera systems provide richer contextual information for industrial anomaly detection. However, traditional methods process each view independently, disregarding the complementary information across viewpoints. Existing multi-view anomaly detection approaches typically employ data-driven cross-view attention for feature fusion but fail to leverage the unique geometric properties of multi-camera setups. In this work, we introduce an epipolar geometry-constrained attention module to guide cross-view fusion, ensuring more effective information aggregation. To further enhance the potential of cross-view attention, we propose a pretraining strategy inspired by memory bank-based anomaly detection. This approach encourages normal feature representations to form multiple local clusters and incorporate multi-view aware negative sample synthesis to regularize pretraining. We demonstrate that our epipolar guided multi-view anomaly detection framework outperforms existing methods on the state-of-the-art multi-view anomaly detection dataset.

> **Extracted A:** What is the role of epipolar geometry-constrained attention in improving cross-view fusion for multi-view anomaly detection?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h064 C type: inquiry_change

arXiv 2508.00471 · 2025-08-01

Semantic and Temporal Integration in Latent Diffusion Space for High-Fidelity Video Super-Resolution

Recent advancements in video super-resolution (VSR) models have demonstrated impressive results in enhancing low-resolution videos. However, due to limitations in adequately controlling the generation process, achieving high fidelity alignment with the low-resolution input while maintaining temporal consistency across frames remains a significant challenge. In this work, we propose Semantic and Temporal Guided Video Super-Resolution (SeTe-VSR), a novel approach that incorporates both semantic and temporal-spatio guidance in the latent diffusion space to address these challenges. By incorporating high-level semantic information and integrating spatial and temporal information, our approach achieves a seamless balance between recovering intricate details and ensuring temporal coherence. Our method not only preserves high-reality visual content but also significantly enhances fidelity. Extensive experiments demonstrate that SeTe-VSR outperforms existing methods in terms of detail recovery and perceptual quality, highlighting its effectiveness for complex video super-resolution tasks.

> **Extracted A:** What is the proposed approach for improving high-fidelity video super-resolution using semantic and temporal-guided information?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

---

broader A question that contains A as a special case.

---

h065 C type: direction_flip

arXiv 2510.22359 · 2025-10-25

EndoSfM3D: Learning to 3D Reconstruct Any Endoscopic Surgery Scene using Self-supervised Foundation Model

3D reconstruction of endoscopic surgery scenes plays a vital role in enhancing scene perception, enabling AR visualization, and supporting context-aware decision-making in image-guided surgery. A critical yet challenging step in this process is the accurate estimation of the endoscope's intrinsic parameters. In real surgical settings, intrinsic calibration is hindered by sterility constraints and the use of specialized endoscopes with continuous zoom and telescope rotation. Most existing methods for endoscopic 3D reconstruction do not estimate intrinsic parameters, limiting their effectiveness for accurate and reliable reconstruction. In this paper, we integrate intrinsic parameter estimation into a self-supervised monocular depth estimation framework by adapting the Depth Anything V2 (DA2) model for joint depth, pose, and intrinsics prediction. We introduce an attention-based pose network and a Weight-Decomposed Low-Rank Adaptation (DoRA) strategy for efficient fine-tuning of DA2. Our method is validated on the SCARED and C3VD public datasets, demonstrating superior performance compared to recent state-of-the-art approaches in self-supervised monocular depth estimation and 3D reconstruction. Code and model weights can be found in project repository: https://github.com/MOYF-beta/EndoSfM3D.

> **Extracted A:** What is the proposed approach for integrating intrinsic parameter estimation into a self-supervised monocular depth estimation framework for endoscopic 3D reconstruction?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h066 C type: role_swap

arXiv 2506.14842 · 2025-06-16

PictSure: Pretraining Embeddings Matters for In-Context Learning Image Classifiers

Building image classification models remains cumbersome in data-scarce domains, where collecting large labeled datasets is impractical. In-context learning (ICL) is a promising paradigm for few-shot image classification (FSIC), but prior work has underexplored the relative importance of encoder pretraining versus fusion-layer training data. We present PictSure, a vision-only ICL family of models that demonstrates the potential of easy-to-use fusion transformer architectures, as well as the need for better embedding representations across a wider range of image domains. In both in-domain and out-of-domain evaluations, we find that representation quality induced by pretraining strongly correlates with downstream ICL performance. Crucially, varying the training dataset for the fusion transformer, from ImageNet alone to diverse multi-domain mixtures, provides limited additional performance gains under the evaluated settings, demonstrating that the fusion layer appears capable of adapting effectively once embeddings are sufficiently structured. These results show that the bottleneck in visual ICL is representation quality, not fusion-module training diversity. To facilitate adoption and reproducibility, we release all model weights as open-source artifacts and provide an MCP server that exposes PictSure as a callable tool for LLM-based agentic systems, enabling few-shot image classification to be invoked directly within AI pipelines without integration overhead. Code can be found at https://github.com/PictSure and models at https://huggingface.co/pictsure.

> **Extracted A:** What factors contribute to the effectiveness of pretraining embeddings in building image classification models for in-context learning?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h067 C type: constraint_flip

arXiv 2504.08638 · 2025-04-11

Transformer Learns Optimal Variable Selection in Group-Sparse Classification

Transformers have demonstrated remarkable success across various applications. However, the success of transformers have not been understood in theory. In this work, we give a case study of how transformers can be trained to learn a classic statistical model with "group sparsity", where the input variables form multiple groups, and the label only depends on the variables from one of the groups. We theoretically demonstrate that, a one-layer transformer trained by gradient descent can correctly leverage the attention mechanism to select variables, disregarding irrelevant ones and focusing on those beneficial for classification. We also demonstrate that a well-pretrained one-layer transformer can be adapted to new downstream tasks to achieve good prediction accuracy with a limited number of samples. Our study sheds light on how transformers effectively learn structured data.

> **Extracted A:** How does a one-layer transformer adapt to new downstream tasks?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h068 C type: goal_swap

arXiv 2502.13891 · 2025-02-19

Highly Dynamic and Flexible Spatio-Temporal Spectrum Management with AI-Driven O-RAN: A Multi-Granularity Marketplace Framework

Current spectrum-sharing frameworks struggle with adaptability, often being either static or insufficiently dynamic. They primarily emphasize temporal sharing while overlooking spatial and spectral dimensions. We propose an adaptive, AI-driven spectrum-sharing framework within the O-RAN architecture, integrating discriminative and generative AI (GenAI) to forecast spectrum needs across multiple timescales and spatial granularities. A marketplace model, managed by an authorized spectrum broker, enables operators to trade spectrum dynamically, balancing static assignments with real-time trading. GenAI enhances traffic prediction, spectrum estimation, and allocation, optimizing utilization while reducing costs. This modular, flexible approach fosters operator collaboration, maximizing efficiency and revenue. A key research challenge is refining allocation granularity and spatio-temporal dynamics beyond existing models.

> **Extracted A:** What is the proposed multi-granularity marketplace framework for adaptive AI-driven spectrum management in the O-RAN architecture?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h069 C type: inquiry_change ★ make broader LONGER than narrower

arXiv 2507.16557 · 2025-07-22

Exploring Gender Bias in Large Language Models: An In-depth Dive into the German Language

In recent years, various methods have been proposed to evaluate gender bias in large language models (LLMs). A key challenge lies in the transferability of bias measurement methods initially developed for the English language when applied to other languages. This work aims to contribute to this research strand by presenting five German datasets for gender bias evaluation in LLMs. The datasets are grounded in well-established concepts of gender bias and are accessible through multiple methodologies. Our findings, reported for eight multilingual LLM models, reveal unique challenges associated with gender bias in German, including the ambiguous interpretation of male occupational terms and the influence of seemingly neutral nouns on gender perception. This work contributes to the understanding of gender bias in LLMs across languages and underscores the necessity for tailored evaluation frameworks.

> **Extracted A:** What are the unique challenges associated with gender bias in German, particularly in relation to ambiguous interpretations of male occupational terms and the influence of seemingly neutral nouns?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h070 C type: direction_flip

arXiv 2502.14499 · 2025-02-20

MLGym: A New Framework and Benchmark for Advancing AI Research Agents

We introduce Meta MLGym and MLGym-Bench, a new framework and benchmark for evaluating and developing LLM agents on AI research tasks. This is the first Gym environment for machine learning (ML) tasks, enabling research on reinforcement learning (RL) algorithms for training such agents. MLGym-bench consists of 13 diverse and open-ended AI research tasks from diverse domains such as computer vision, natural language processing, reinforcement learning, and game theory. Solving these tasks requires real-world AI research skills such as generating new ideas and hypotheses, creating and processing data, implementing ML methods, training models, running experiments, analyzing the results, and iterating through this process to improve on a given task. We evaluate a number of frontier large language models (LLMs) on our benchmarks such as Claude-3.5-Sonnet, Llama-3.1 405B, GPT-4o, o1-preview, and Gemini-1.5 Pro. Our MLGym framework makes it easy to add new tasks, integrate and evaluate models or agents, generate synthetic data at scale, as well as develop new learning algorithms for training agents on AI research tasks. We find that current frontier models can improve on the given baselines, usually by finding better hyperparameters, but do not generate novel hypotheses, algorithms, architectures, or substantial improvements. We open-source our framework and benchmark to facilitate future research in advancing the AI research capabilities of LLM agents.

> **Extracted A:** What are the challenges and opportunities for advancing the AI research capabilities of LLM agents using the proposed framework and benchmark?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

---

narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

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h071 C type: role_swap

arXiv 2510.07731 · 2025-10-09

oMeBench: Towards Robust Benchmarking of LLMs in Organic Mechanism Elucidation and Reasoning

Organic reaction mechanisms describe the step-wise elementary processes by which reactants transform into intermediates and products, and are fundamental to understanding chemical reactivity and guiding molecular and reaction de-sign. While large language models (LLMs) have shown promise on chemical tasks such as synthesis design, it remains unclear to what extent this reflects genuine chemical reasoning capabilities: the ability to generate chemically valid intermediates, maintain consistency across reaction steps, and follow logically coherent multi-step pathways. To investigate this, we introduce oMeBench, the first large-scale, expert-curated benchmark for organic mechanism reasoning, comprising over 10,000 annotated mechanistic steps with reaction type labels, intermediate structures, and difficulty ratings. To enable fine-grained evaluation, we further propose oMeS, a dynamic scoring framework that jointly assesses step-level logical consistency and chemical structural similarity. Systematic evaluation of state-of-the-art LLMs reveals that while current models exhibit promising chemical intuition, they struggle to produce correct and consistent reasoning across multi-step mechanisms. Notably, combining prompting strategies with fine-tuning enables smaller-scale models to achieve performance comparable to closed-source frontier models. We hope oMeBench will serve as a rigorous foundation for advancing AI systems toward genuine chemical reasoning.

> **Extracted A:** What is the impact of incorporating prompting strategies with fine-tuning on the performance of small-scale models compared to closed-source frontier models?

C type: swap two roles: "how does X affect Y" → "how does Y affect X"

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h072 C type: constraint_flip

arXiv 2510.14738 · 2025-10-16

AutoRubric: Rubric-Based Generative Rewards for Faithful Multimodal Reasoning

Multimodal large language models (MLLMs) have rapidly advanced from perception tasks to complex multi-step reasoning, yet reinforcement learning with verifiable rewards (RLVR) often leads to spurious reasoning since only the final-answer correctness is rewarded. To address this limitation, we propose AutoRubric, a framework that integrates RLVR with process-level supervision through automatically collected rubric-based generative rewards. Our key innovation lies in a scalable self-aggregation method that distills consistent reasoning checkpoints from successful trajectories, enabling problem-specific rubric construction without human annotation or stronger teacher models. By jointly leveraging rubric-based and outcome rewards, AutoRubric achieves state-of-the-art performance on six multimodal reasoning benchmarks and substantially improves reasoning faithfulness in dedicated evaluations.

Extracted A: What is the key innovation of AutoRubric in integrating RLVR with process-level supervision?

C type: flip a constraint: "without labels" → "with labels", "on CPUs" → "on GPUs"

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain…).
broader	A question that contains A as a special case.

h073 C type: goal_swap

arXiv 2505.17660 · 2025-05-23

DAM-GT: Dual Positional Encoding-Based Attention Masking Graph Transformer for Node Classification

Neighborhood-aware tokenized graph Transformers have recently shown great potential for node classification tasks. Despite their effectiveness, our in-depth analysis of neighborhood tokens reveals two critical limitations in the existing paradigm. First, current neighborhood token generation methods fail to adequately capture attribute correlations within a neighborhood. Second, the conventional self-attention mechanism suffers from attention diversion when processing neighborhood tokens, where high-hop neighborhoods receive disproportionate focus, severely disrupting information interactions between the target node and its neighborhood tokens. To address these challenges, we propose DAM-GT, Dual positional encoding-based Attention Masking graph Transformer. DAM-GT introduces a novel dual positional encoding scheme that incorporates attribute-aware encoding via an attribute clustering strategy, effectively preserving node correlations in both topological and attribute spaces. In addition, DAM-GT formulates a new attention mechanism with a simple yet effective masking strategy to guide interactions between target nodes and their neighborhood tokens, overcoming the issue of attention diversion. Extensive experiments on various graphs with different homophily levels as well as different scales demonstrate that DAM-GT consistently outperforms state-of-the-art methods in node classification tasks.

> **Extracted A:** What is the proposed method for addressing the limitations of neighborhood token generation and attention mechanism in node classification tasks?

C type: change the goal: accuracy → speed, robustness → fairness

A (edited)	Edit the extracted question until it is faithful to the paper. If it is already right, copy it.
B	Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).
B_min	Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).
C	Near-miss of the type shown above: almost the same words, but a DIFFERENT question.
C_tilde	C's meaning in different words: Jaccard with A ≤ 0.3.
narrower	A special case of A (add a dataset, model, domain...).
broader	A question that contains A as a special case.

h074 C type: inquiry_change

arXiv 2503.19011 · 2025-03-24

RomanTex: Decoupling 3D-aware Rotary Positional Embedded Multi-Attention Network for Texture Synthesis

Painting textures for existing geometries is a critical yet labor-intensive process in 3D asset generation. Recent advancements in text-to-image (T2I) models have led to significant progress in texture generation. Most existing research approaches this task by first generating images in 2D spaces using image diffusion models, followed by a texture baking process to achieve UV texture. However, these methods often struggle to produce high-quality textures due to inconsistencies among the generated multi-view images, resulting in seams and ghosting artifacts. In contrast, 3D-based texture synthesis methods aim to address these inconsistencies, but they often neglect 2D diffusion model priors, making them challenging to apply to real-world objects To overcome these limitations, we propose RomanTex, a multiview-based texture generation framework that integrates a multi-attention network with an underlying 3D representation, facilitated by our novel 3D-aware Rotary Positional Embedding. Additionally, we incorporate a decoupling characteristic in the multi-attention block to enhance the model's robustness in image-to-texture task, enabling semantically-correct back-view synthesis. Furthermore, we introduce a geometry-related Classifier-Free Guidance (CFG) mechanism to further improve the alignment with both geometries and images. Quantitative and qualitative evaluations, along with comprehensive user studies, demonstrate that our method achieves state-of-the-art results in texture quality and consistency.

> **Extracted A:** What is the proposed approach for integrating a multi-attention network with a 3D representation to enhance texture generation from 2D images?

C type: change the kind of question: "how does" → "why does" / "can we" / "to what extent"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

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narrower A special case of A (add a dataset, model, domain...).

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broader A question that contains A as a special case.

h075 C type: direction_flip ★ make broader LONGER than narrower

arXiv 2508.06082 · 2025-08-08

SwiftVideo: A Unified Framework for Few-Step Video Generation through Trajectory-Distribution Alignment

Diffusion-based or flow-based models have achieved significant progress in video synthesis but require multiple iterative sampling steps, which incurs substantial computational overhead. While many distillation methods that are solely based on trajectory-preserving or distribution-matching have been developed to accelerate video generation models, these approaches often suffer from performance breakdown or increased artifacts under few-step settings. To address these limitations, we propose \textbf{\emph{SwiftVideo}}, a unified and stable distillation framework that combines the advantages of trajectory-preserving and distribution-matching strategies. Our approach introduces continuous-time consistency distillation to ensure precise preservation of ODE trajectories. Subsequently, we propose a dual-perspective alignment that includes distribution alignment between synthetic and real data along with trajectory alignment across different inference steps. Our method maintains high-quality video generation while substantially reducing the number of inference steps. Quantitative evaluations on the OpenVid-1M benchmark demonstrate that our method significantly outperforms existing approaches in few-step video generation.

Extracted A: What is the proposed SwiftVideo framework for generating few-step videos?

C type: flip the direction: "improve" → "harm", "increase" → "reduce"

A (edited) Edit the extracted question until it is faithful to the paper. If it is already right, copy it.

B Same question, different words. Share at most 30 % of A's words (Jaccard ≤ 0.3).

B_min Same question, tiny edit: keep at least 75 % of A's words (e.g. change one verb).

C Near-miss of the type shown above: almost the same words, but a DIFFERENT question.

C_tilde C's meaning in different words: Jaccard with A ≤ 0.3.

narrower A special case of A (add a dataset, model, domain...).

broader A question that contains A as a special case.